NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

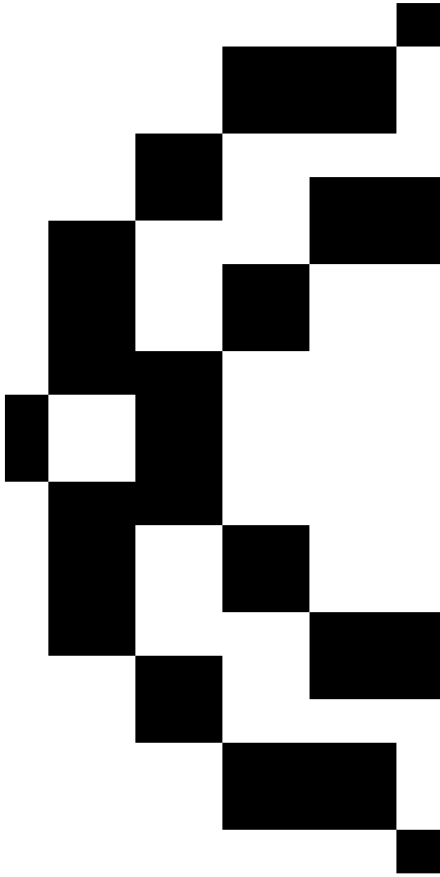
Practicals Week 07 - Lists & For Loops

Exercise 1

Gertrude has a lot of trouble remembering what she needs to buy. Ask her how many items she intends to buy, then ask her for each item's name in turn (exactly that many times) and store the names in a list. Print the list so Gertrude can use it as a reference.

Example

- `Input:  amount = 3, items = "milk", "bread", "eggs"`
- `Output: ['milk', 'bread', 'eggs']`

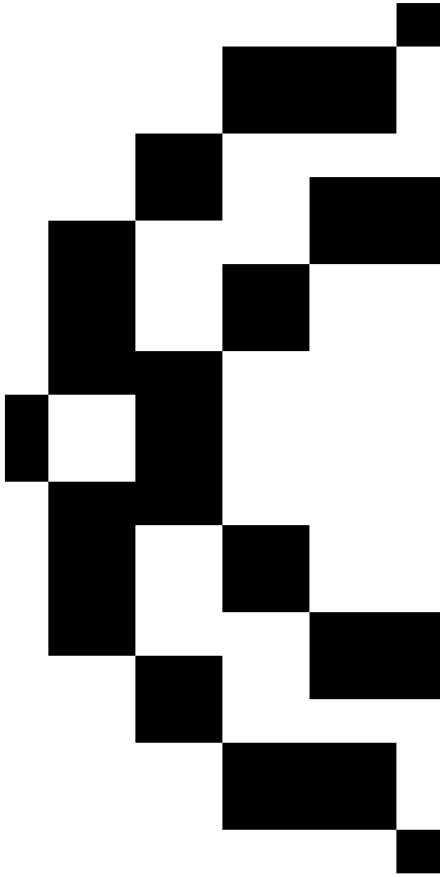


Exercise 2

Create a list of your choosing. Multiply the values at odd positions by 2 and the values at even positions by 3. Treat index 0 as an even position.

Example

- `Input:  my_list = [1, 2, 3, 4, 5, 6]`
- `Output: [3, 4, 9, 8, 15, 12]`

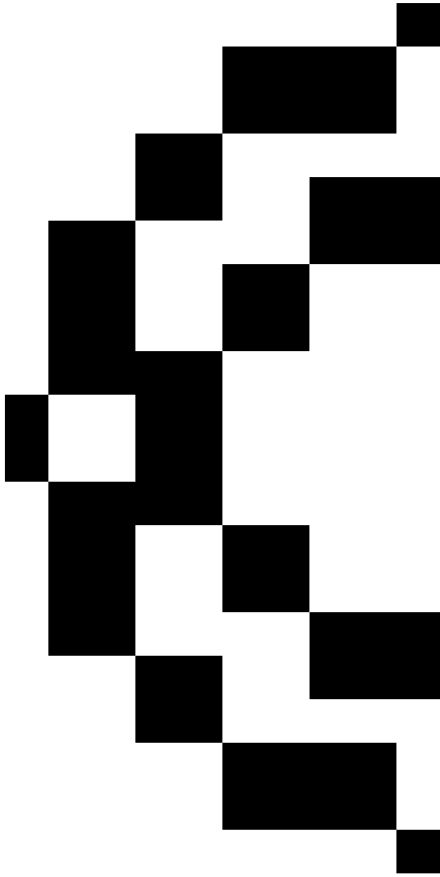


Exercise 3

Ask the user how many names they have. Create an empty list called `names` and fill it by asking the user for each name in turn, storing the answers in the list. Finally, print a greeting using the user's first and last name.

Example

- `Input:  n = 3, names = "Maria", "Carla", "Silva"`
- `Output: nice to meet you Maria Silva`

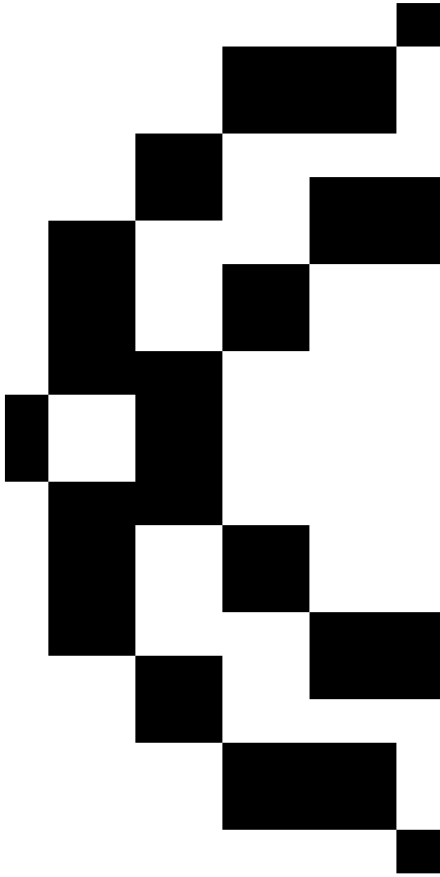


Exercise 4

Define a list with four positive integers of your choosing. Change the first element of the list to `100`. Then, using a `for` loop, calculate the sum of all elements that are bigger than 6.

Example

- `Input:  my_list = [1, 2, 30, 4]`
- `Output: 130`

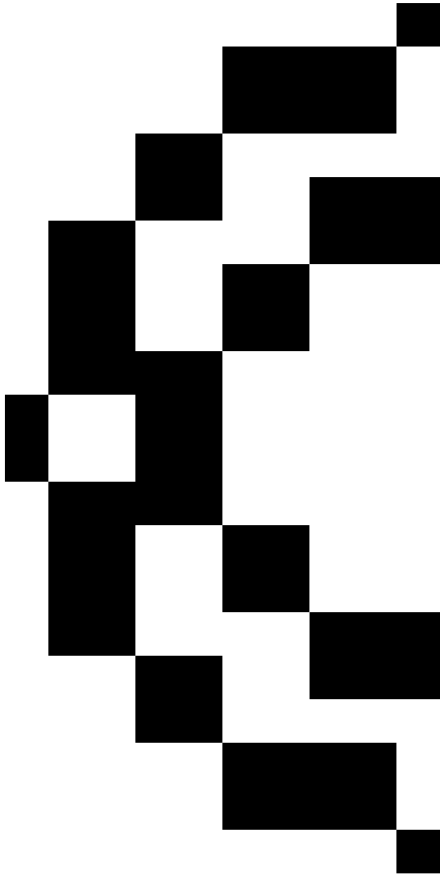


Exercise 5

Ask the user how many numbers they have, then build a list of that many integers using the same method as previous exercises. Count how many of the numbers are even and print the amount, then print how many are odd.

Example

- `Input:  amount = 5, numbers = 2, 3, 4, 5, 6`
- `Output: There are 3 even numbers.`
- `        There are 2 odd numbers.`

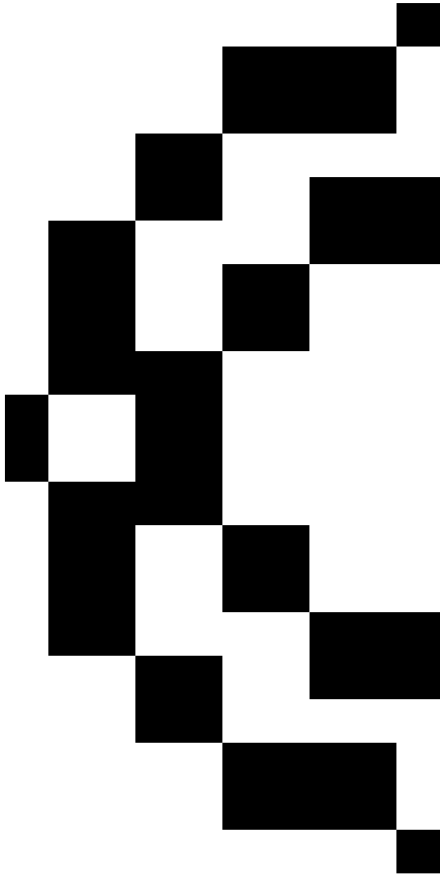


Exercise 6

Ask Gertrude how many games she bought this month, then ask her for the price of each one and store the prices in a list. Calculate and print how much she spent in total. If she spent more than €100, let her know she has a problem.

Example

```
Input:  amount = 2, prices = 45.0, 70.0
Output: Gertie spent 115.0 euros in games
        Gertie has issues
```

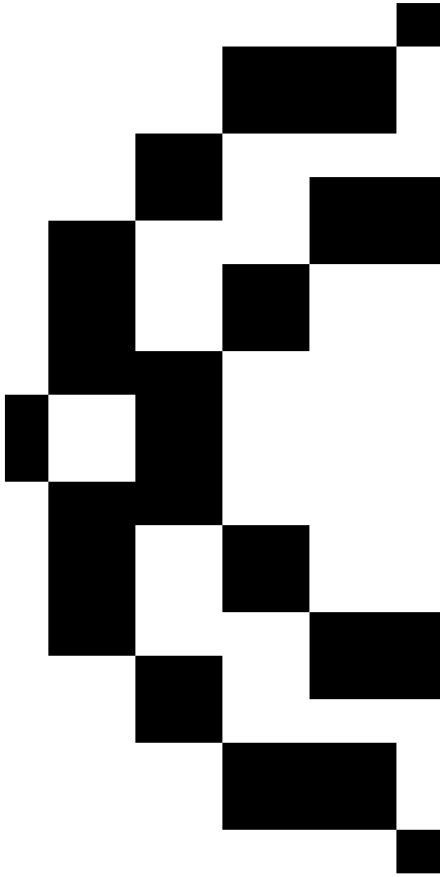


Exercise 7

Ask Gertrude how many games she bought last month, then ask her for the name of each game and store the names in a list. Print how many *Assassin's Creed* games she bought. Note that there are many Assassin's Creed games, and all of them contain "Assassin's Creed" somewhere in their title (e.g. "Assassin's Creed I", "Assassin's Creed Revelations").

Example

- `Input:  amount = 3, games = "Assassin's Creed II", "Tetris", "Assassin's Creed Origins"`
- `Output: Gertie bought 2 assassin's creed games.`

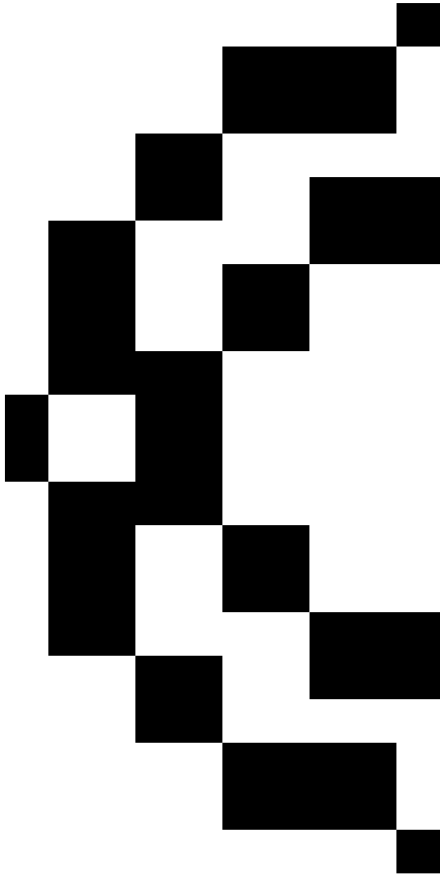


Exercise 8

Ask the user how many children they would like to have, then ask for the name they would like to give each child and store the names in a list. Print a list containing the number of letters in each name, in the same order.

Example

- `Input:  n = 3, names = "John", "Johnna", "Ja"`
- `Output: [4, 6, 2]`

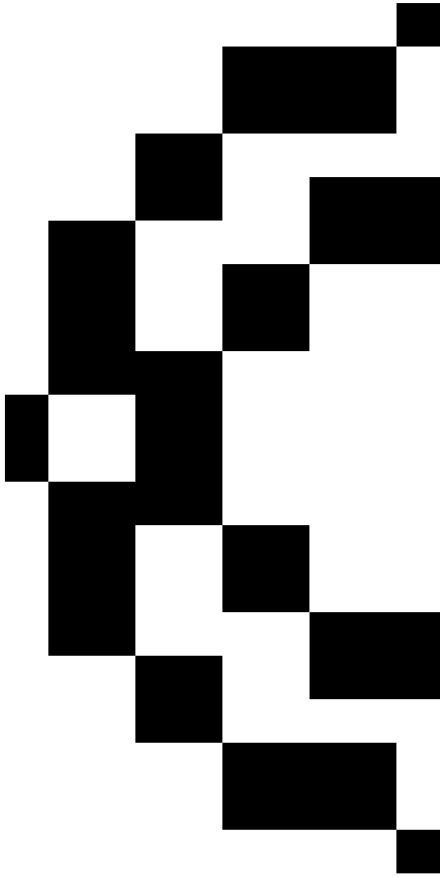


Exercise 9

Modify the previous exercise: ask the user how many children they would like to have and what their names are, then print how many of the given names begin with a vowel.

Example

- `Input:  n = 3, names = "Emma", "John", "Ivy"`
- `Output: 2 names begin with a vowel`



NOVA IMS INFORMATION MANAGEMENT SCHOOL

Thank you

Acreditações e Certificações da NOVA IMS

Cofinanciado por